

 HelioSketch
Solar Design Report

8

Panels

3.2

kW DC

4,807

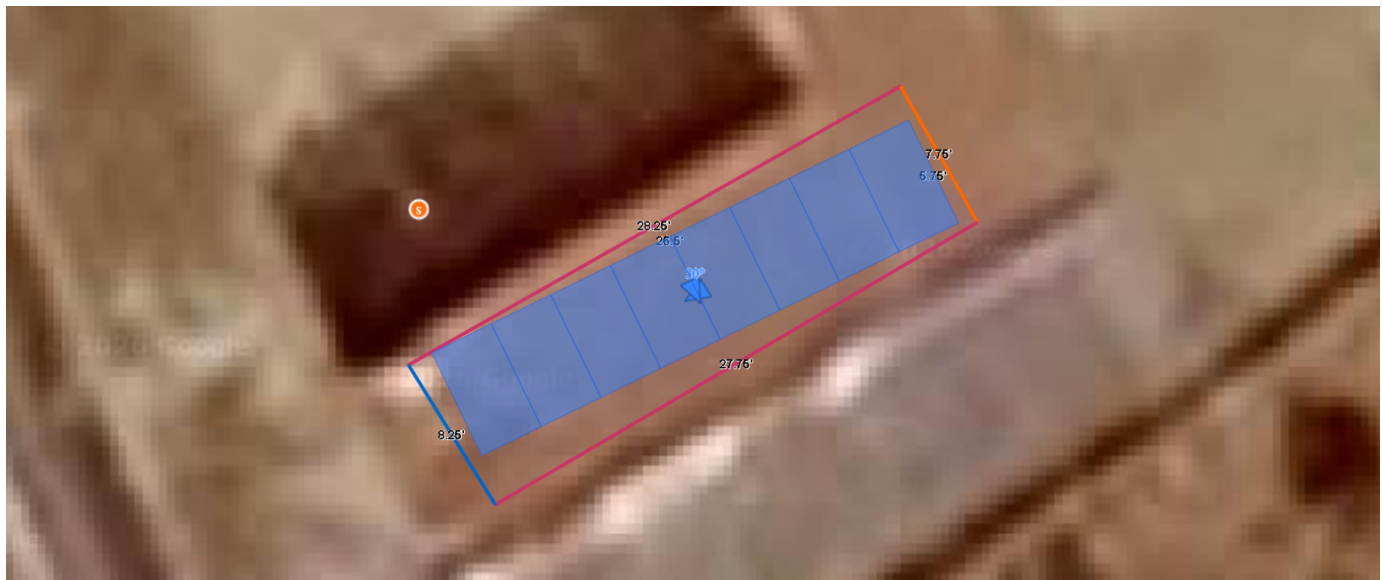
kWh/yr

5

kWh Storage

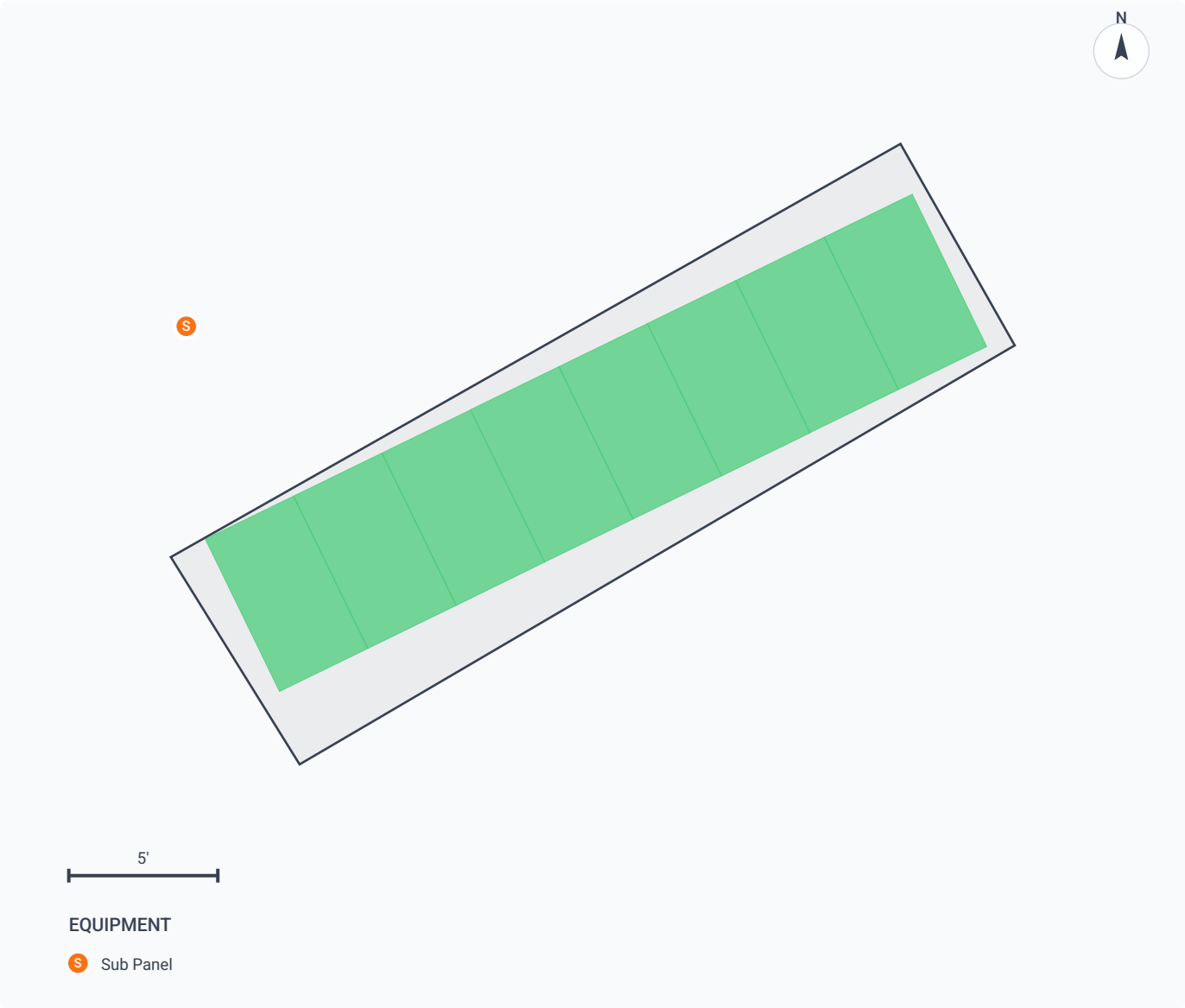
Fazouane
Ahfir Oriental Morocco

September 30, 2026
Prepared by Chikhi ayoub
ayoubchikhi2004@gmail.com



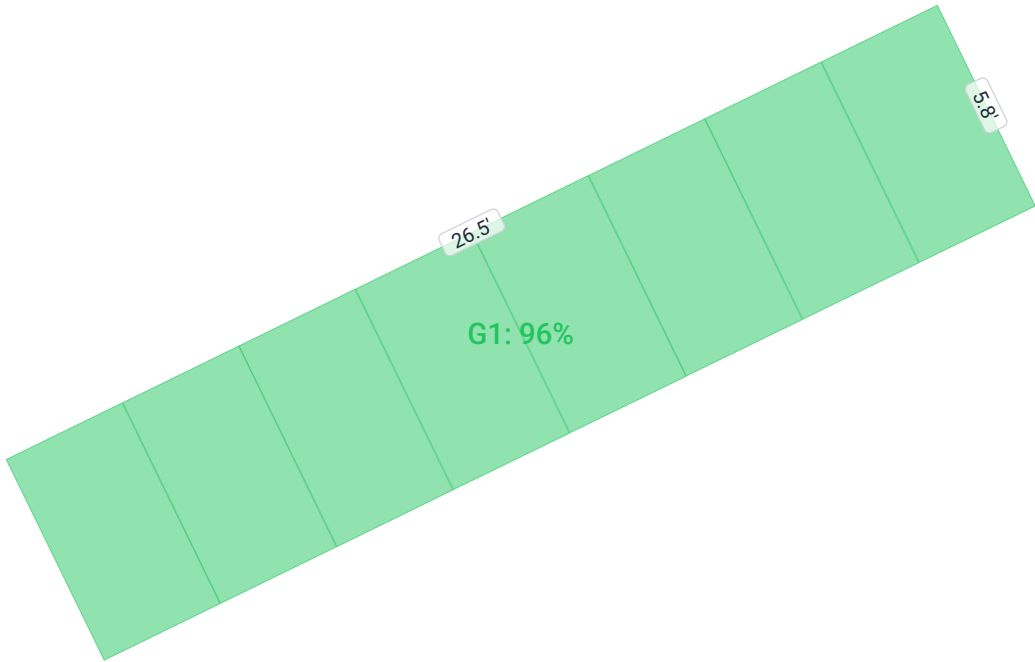
Site Layout

Ahfir Oriental Morocco



Ground Mount Layout

Ground-mounted array layout with tilt/orientation factor



Solar Production Factor (TOF)

The Tilt & Orientation Factor indicates how well each ground mount array captures solar energy relative to the optimal south-facing orientation.

0–50% Poor	50–65% Fair	65–75% Good	75–85% Very Good	85–100% Excellent	
Array		TOF	Rating	Orientation	Panels
G1		96%	Excellent	154° SSE	8

System Overview

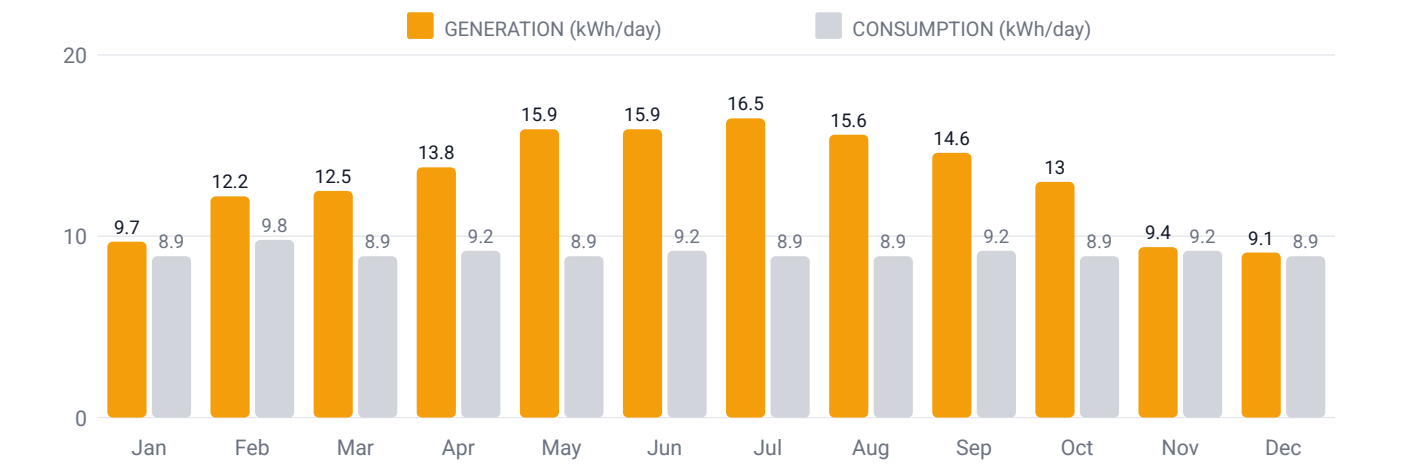
Ahfir Oriental Morocco

System Details	
Panel Module	ALEXUS ALEX-405-B-54-S
Total Panels	8
Panel Wattage	405 W
Panel Dimensions	67.8" x 44.6"
System Size (DC)	3.2 kW
Number of Arrays	1
Est. Annual Production	4,807 kWh
Solar Offset	146% of annual consumption
Production Ratio	1484 kWh/kW
Sized for Future Loads	Mini-split AC unit, Electric water heater (+3,300 kWh/yr)
Battery Storage	5 kWh

Array Details

#	Type	Panels	Capacity	Tilt	Azimuth	Area	TOF
G1	Ground Mount	8	3.2 kW	30°	154° SSE	173 ft²	96% Excellent

System Performance



Annual Production: 4,807 kWh | Annual Consumption: 3,300 kWh | Solar Offset: 146% (+1,507 kWh)

Usage shown includes your planned future loads: Mini-split AC unit, Electric water heater (+3,300 kWh/yr on top of today's bill).

Production estimates are based on typical weather patterns and system specifications. Actual production may vary due to weather, shading, soiling, equipment degradation, and other factors.

Project Investment

Ahfir Oriental Morocco

TOTAL SYSTEM COST

€14.840

€4.58/W

3.2 kW DC

Category	Description	Total
Solar Panels	8 panels × 405W	€2.722
Inverter / MLPE	Inverter, optimizers, monitoring	€1.928
Racking	Rails, mounts, clamps, flashing	€1.588
Balance of System	Wire, conduit, disconnects, connectors	€1.701
Procurement	Shipping, handling, logistics	€794
Labor & Install	Installation, commissioning, permitting	€2.608
Solar subtotal		€11.340
Battery Storage	5 kWh storage	€3.500
Total Project Cost		€4.58/W €14.840

Pricing is an estimate based on system size and standard installation costs. Final pricing may vary based on site conditions, equipment selection, permitting requirements, and applicable incentives or rebates. Does not include sales tax unless noted.

Battery Storage Simulation

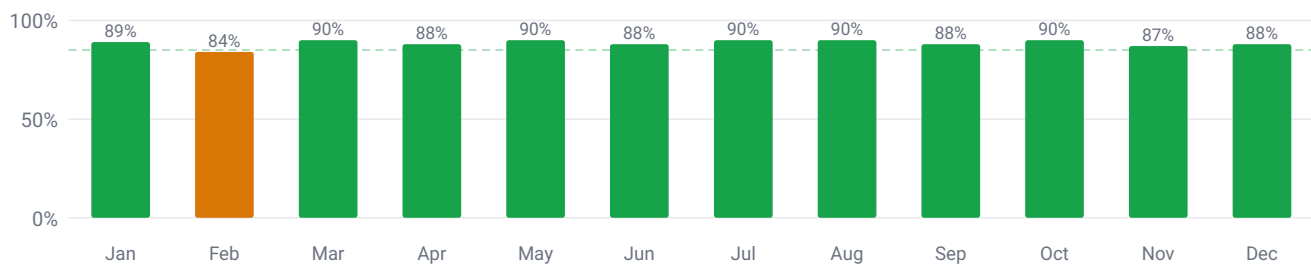
Load profile: Caribbean Residential (night fans/AC)

Daily Energy Balance

Battery Capacity	5 kWh
Usable Capacity	5.0 kWh
Round-Trip Efficiency	90%
Daily Production	14.2 kWh
Daily Consumption	9.0 kWh
Grid Import	1.0 kWh
Excess Solar (curtailed)	5.6 kWh
Battery Contribution	4.7 kWh
Self-Consumption (% of solar used on-site)	65%
Self-Sufficiency (% of load met by solar+battery)	89%

Monthly Self-Sufficiency

Percentage of monthly load met by solar + battery (vs. grid). Dashed line marks the 85% target.



Worst case: An overcast Feb day pulls roughly 4.3 kWh from the grid (~43% of that day's load).

Sizing Assessment

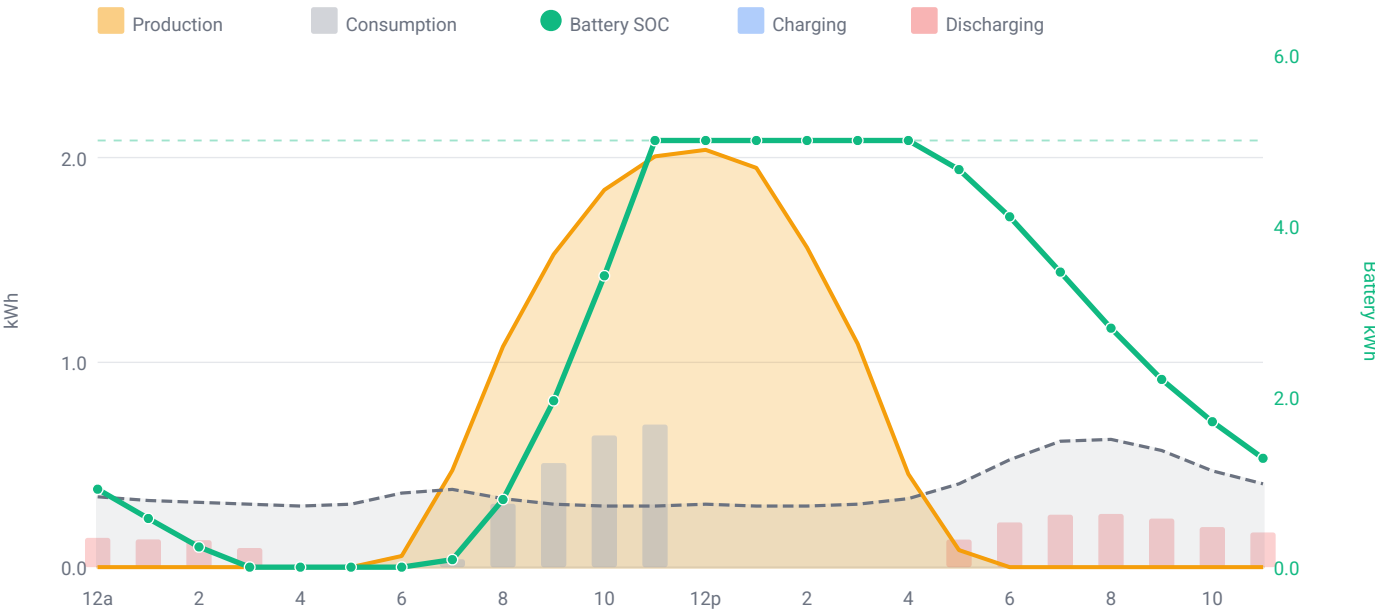
Solar Oversized

Industry targets: Self-Consumption $\geq 75\%$, Self-Sufficiency $\geq 85\%$.

Recommended Solar Size	2.3 kW	Current: 3.2 kW
Recommended Battery (usable)	7.3 kWh	Current: 5.0 kWh

Significant solar export. Reduce panel count (~28%), add battery capacity, or add daytime electrified loads (EV, heat pump, pool pump).

Daily Energy Profile & Battery Simulation



Simulation based on typical residential load profile and average daily production. Actual performance will vary with real consumption patterns, weather, and equipment.

Bill of Materials

Major Equipment

Item	Description	Qty	Unit
Solar Panel	ALEXUS ALEX-405-B-54-S, 405W	8	pcs
Battery	5 kWh battery	1	pcs

Quantities are estimates based on design layout. Actual quantities may vary based on site conditions and specific product selection.